

Pranav Manjunath

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Rhodes Graduate Fellowship Awardee

PhD candidate passionate about transforming multi-modal data into insights, provide innovative solutions to complex real-world healthcare problems, build productive cross functional collaborations and lead by example.

EDUCATION

PhD in Biomedical Engineering <i>Duke University (Advisor: Dr. Timothy Dunn)</i>	Aug 2022 – Present <i>Durham, NC</i>
MS in Interdisciplinary Data Science (GPA: 3.95) <i>Duke University</i>	Aug 2020 – May 2022 <i>Durham, NC</i>
BE in Computer Science and Engineering (Certificate of Distinction) <i>Visvesvaraya Technological University (PESIT)</i>	Aug 2016 – May 2020 <i>Bangalore, India</i>

EXPERIENCE

Visiting Researcher <i>Human-Computer Interaction Institute, Carnegie Mellon University</i> Collaborating with Dr. Adam Perer in the Data Interactions Group at CMU on evaluating and studying how multimodal AI systems can support clinical stakeholder and augment traumatic brain injury care.	Oct 2025 – Present <i>Pittsburgh, PA</i>
Co-leader - Discovering AI Applications for Traumatic Brain Injury Care <i>Duke University</i> - Co-initiated and leading the Bass Connections project, securing over \$30,000 in competitive funding to conduct the largest qualitative/mixed-methods research study focused on identifying AI-driven opportunities to improve TBI care at Duke. <i>Project Website</i> - Collaborated with senior faculty across multiple departments to design and teach the project-based course to an interdisciplinary team of over 20 students. - Awarded the Rhodes Graduate Fellowship for Interdisciplinary Research for spearheading this project - Students present this research at the National Collegiate Research Conference 2026 at Harvard, Society-Centered AI Conference 2026, Duke National Health Research Conference 2026, and the Duke AI Health Data Science Showcase 2025.	Aug 2024 – May 2025 <i>Durham, NC</i>
Co-leader - Graduate School Committee: DukeREP <i>Duke University</i> - Outreach program aiming increase diversity in STEM fields by encouraging high-school students to pursue academic and professional careers in STEM - Spearheaded efforts to recruit mentors and instructors within the PhD program to support and guide high school students through teaching and mentoring initiatives.	May 2023 – Present <i>Durham, NC</i>
Health AI Researcher - +DS Program <i>Duke University</i> Worked with Prof. Michael Pencina, Prof. Ricardo Henao and a team of Duke researchers on a NIH grant project to build a stroke survival predictive model.	May 2021 – May 2022 <i>Durham, NC</i>
Capstone Researcher - ViacomCBS/MTV <i>Duke University</i> - Worked with MTV to find solutions to increase the voter turnout rates amongst youth in the US elections using geospatial data. Tested a hypothesis that proximity of polling location to universities has an inverse correlation with the student voter turnout rate. - Won Silver Distinction at Anthem Awards 2023 under the category Awareness Media (for profit). - Project Findings and Insights featured in <i>Business Wire</i> and <i>USA Today</i> - helping MTV to start social campaigns to improve student voter access.	Aug 2021 – May 2022 <i>Durham, NC</i>
Co President - Duke Interdisciplinary Product Management Club <i>Duke University</i>	May 2021 – May 2022 <i>Durham, NC</i>

- Co-founder of a club at Duke to bridge Product Management, Data Science, and Machine Learning. Main responsibilities: Organize technical workshops, seminars, datathons, productions, strategic case competitions

Project Manager - Data+

May 2021 – Aug 2021

Duke University

Durham, NC

- Managed a team of Duke undergraduate students to develop machine learning features that can be used to identify unknown web attacks.
- Outcome incorporated into Duke's IT security infrastructure to help protect the network.

Data Scientist Summer Intern

May 2021 – Aug 2021

Advance Auto Parts

Raleigh, NC

- Clustering and topic modelling customer product reviews into key themes utilizing NLP unsupervised algorithms. Used by business leaders to address customer pain points and trends
- Feature engineered product, customer, location and vehicle data from various sources leveraging SQL and Python
- Presented results and recommendations to stakeholders, company technical interest group and C-suite executives.

Graduate Teaching Assistant

Dec 2020 – Present

Duke University

Durham, NC

- **Introduction to Data Science - Dr. Timothy Dunn:** Organize weekly office hours to review and teach course material and grade student assignments.
- **Design Health 3 - Dr. Eric Richardson, Dr. Paul Fearis, Dr. Joseph Knight:** Organize, create optimal class schedules templates and provide feedback and grade students on their presentations.
- **Programming for Data Analytics (Fuqua School of Business) - Prof. Brian Cozzi:** Organize weekly office hours to help students with coursework and teach them concepts in Python and R programming.
- **Data Engineering in the Cloud - Prof. Noah Gift:** Help structure the course, grade and review assignments, organize office hours to help students with the coursework.
- **Introduction to Machine Learning - Prof. Kyle Bradbury:** Help grade and review assignments, organize office hours to help students with the coursework.
- **Data Analytics and Applications (Fuqua School of Business) - Prof. Mattia Ciollaro:** Organize weekly office hours to help students with coursework and teach them concepts in Machine Learning.
- **Cloud Computing for Data Analysis - Prof. Noah Gift:** Help structure the course, grade and review assignments, organize office hours to help students with the coursework.
- **Python Winter Course and MIDS Bootcamp - Prof. Andrew Hilton and Prof. Genevieve Lipp:** Organize office hours for students to help teach them Python programming.

PUBLICATIONS

How Much Information Is Enough? A Design Space for Mediating Personalized Information with AI in Emergency Care 2026

Pranav Manjunath, Tolu Oyesanya, Adam Perer, Timothy W. Dunn

ACM Conference on Human Factors in Computing Systems (CHI) 2026

Lightning Talk at CHI 2026 Workshop: Toward Relationship-Centered Care with AI

Trust Your Neighbors: Multimodal Patient Retrieval for TBI Prognosis 2025

Pranav Manjunath, Brian Lerner, Timothy Dunn

Journal: IEEE Journal of Biomedical and Health Informatics (JBHI)

Paper Link | Oral Presentation at IEEE BHI 2025 | **Best Paper Award**

Context-Aware Filtering of Unstructured Radiology Reports by Anatomical Region 2025

Zakk Heile, Pranav Manjunath, Brian Lerner, Samuel Berchuck, Monica Agrawal, Timothy W. Dunn

Conference: Machine Learning for Health (ML4H)

Paper Link

Personalized Case- and Evidence-Based TBI Prognosis with Small Language Models 2025

Pranav Manjunath, Syed M. Adil, Benjamin Wissel, Daniel Sexton, Brian Lerner, Timothy Dunn

Conference: IEEE Biomedical Health Informatics (IEEE BHI)

Paper Link | Oral Presentation at IEEE BHI 2025 | **Top 9% of Submissions**

Towards Interactive and Interpretable Image Retrieval-Based Diagnosis: Enhancing Brain Tumor Classification with LLM Explanations and Latent Structure Preservation 2024

Pranav Manjunath, Brian Lerner, Timothy Dunn

Conference: Artificial Intelligence in Medicine (AIME 2024)

Paper Link | Oral Presentation at AIME 2024 | **Best Student Paper Award**

Apriori Algorithm and Decision Tree Classification Methods to Mine Educational Data for Evaluating Graduate Admissions to U.S. Universities 2019

Pranav Manjunath, Kushal Naidu

Book Chapter: Algorithms for Intelligent Systems (Springer)

Paper Link

EyesPro: Protect Your Eyes 2018

Pranav Manjunath, Nimisha V. Arun

Journal: International Journal of Scientific Research in Science and Technology (RISE 2018 Inter-College Research Competition)

Paper Link | **Best Paper Award**

POSTERS & ABSTRACTS

Dual Retrieval for Personalized TBI Prognosis Using Real-World Evidence and Clinical Guidelines 2025

Pranav Manjunath, Syed Adil, Ben Wissel, Daniel Sexton, Brian Lerner, Timothy Dunn

Stanford AI+HEALTH 2025

Accepted Poster

Personalizing Total Knee Arthroplasty: A Deep Learning Approach to Restoring Natural Alignment with Anatomic Key-point Identification 2024

J. Helmkamp, Pranav Manjunath, B. Valan, P. Raghuvanshi, M. Izzi, Timothy Dunn, Seyler

Orthopedic Trauma Association (OTA 2024)

Accepted Abstract - Podium Presentation

AI-Driven Content-Based Image Retrieval to Support Brain Tumor Diagnosis 2023

Pranav Manjunath, Brian Lerner, Timothy Dunn

Conference: Duke AI Health Annual Conference

Accepted Poster

VNA to Z: An Automated Pipeline for Retrieving and Processing Duke Health CT Scans 2023

Brian Lerner, Pranav Manjunath, Timothy Dunn

Conference: Duke AI Health Annual Conference

Accepted Poster

SKILLS

Programming Languages: Python, R, SQL (MySQL and Postgres), C++

Cloud Services: AWS Machine Learning Certified, Google Cloud Platform

EXTRA CURRICULAR ACTIVITIES

Professional Violinist and Music Teacher

Trained in Indian Classical Carnatic Music

Co-founded Boston Brothers, an Indian classical and fusion violin band. Collaborated with international musicians, performed over 600 concerts worldwide, and have received multiple awards, honors and citations.

Secured 1st Rank in Vidwath, the highest Indian classical Carnatic music examination

Taught 25 students at a music academy in India and continues to conduct classes for several students in the US.

Contribute part of my earnings to charity, to support the education of poor students. I have performed for several fund-raising concerts to support local artists during the COVID-19 pandemic. As a band we have played in old age homes, charities, orphanages, to help spread the joy of music.